

Appendix F: The Optimal-Policy Gauntlet — A Negative Result (Final, v3 + v4)

UIP Phase 1 Review Packet · Falsified preregistered claim · Companion to preprint DOI: [10.5281/zenodo.21246247](https://doi.org/10.5281/zenodo.21246247) · July 2026

Code: `appendix_f_optimal_gauntlet.py` at github.com/derekhone/uip-phase1-testbeds

Headline result: FALSIFIED — twice, at the preregistered threshold. The claim under test was that the framework's decision quantities $\{C(A), R, \hat{\gamma}\}$ are near-sufficient statistics for optimal coherence-preserving control (preregistered: within 1% of a near-optimal adversary; kill-condition: gap $> 5\%$). Round 1 (policy v3): relative gap **12.19%**. Round 2, a single preregistered predictive extension (v4, online drift-schedule estimator): gap **22.06%** — worse. The control-sufficiency claim is closed. No further iterations were run; closing the gap requires standard dynamic programming, which contains no framework content.

1. Claim under test and preregistration

Prior appendices tested the framework against *static* baselines (always-correct, never, alternating, random). Appendix C/D/E showed it beats those in the restricted-access regime. The stronger claim — implicit in treating $dC/dt = G - L$ as an operating equation — is that the framework's quantities capture *what matters* for control, i.e., that a policy computed from them approaches the best achievable performance.

Preregistered (before running):

P-F1: the inheritance-guided policy v3 attains mean target coherence within 1% of a near-optimal adversary on the identical noise realization.

Kill-condition: a relative gap $> 5\%$ falsifies the sufficiency claim.

Revision budget: at most one preregistered extension permitted after a failure; no iteration beyond that.

2. The adversary

Restricted-access testbed (Appendix C setup: accessible relay A, shielded target B, alternating dephasing attack, correction = partial re-preparation of A, $T = 90$ cycles, binary act/wait decision per cycle). The adversary is a **beam search** over the full 2^{90} action-sequence space (beam width 200, scored by final-run mean $C(B)$), refined by single-flip hill-climbing to a local optimum. The dynamics are deterministic given the noise schedule, so the adversary's score is a **lower bound** on true optimal performance: any tightening (exact DP, wider beam) can only widen the reported gap and strengthen this negative result.

3. Round 1: v3 vs the adversary

Policy	Mean C(B)	Relative gap vs optimal
Near-optimal (beam search + hill-climb)	0.5257	—
UIP v3 (inheritance-guided)	0.4616	12.19%

P-F1: FAIL — gap 12.19% > 5% kill threshold.

Diagnosis. Inspection of the optimal action sequence shows it is not a smooth threshold policy in any instantaneous quantity: it exploits *anticipatory burst structure*, phase-locked to the noise schedule — clusters of corrections timed just before attack windows shift, pre-loading the relay so refreshed coherence arrives at the target as the attack lands. The framework's quantities $\{C(A), R, \gamma\}$ are instantaneous; they carry no timing/lookahead information and cannot represent this strategy class.

4. Round 2: the single permitted extension (v4)

Preregistered hypothesis: the deficit is *information*, not optimization — if the policy is given an online estimate of the noise schedule (a drift-schedule estimator inferring the attack period and phase from the observed decay history of noisy coherence estimates), an anticipatory version of the threshold policy will close the gap to within 5%.

Policy	Mean C(B)	Relative gap vs optimal
Near-optimal	0.5257	—
UIP v3	0.4616	12.19%
UIP v4 (v3 + schedule estimator + anticipatory bursts)	0.4097	22.06%

P-F2: FAIL — v4 performed *worse* than v3. Knowing *when* the noise strikes is not enough: the optimal burst pattern reflects genuine multi-step lookahead optimization over the coupled transfer/decay dynamics, and a hand-designed anticipatory heuristic layered on threshold logic mistimed its bursts badly enough to underperform the purely reactive policy. The revision budget is spent. **The claim is closed as falsified.**

5. What this result means

- **Scope of the framework, sharpened:** the UIP bookkeeping predicts *boundaries* (where regimes change: crossovers, fixed points — Appendices A, D, E) but does **not** encode enough information to act near-optimally *within* a regime. It is a boundary-predicting method, not a control theory.
- **Why we stopped:** a v5 with richer state (e.g., learned value functions) would close the gap only by becoming standard dynamic programming / reinforcement learning, at which point the framework contributes nothing. Continuing would be curve-fitting toward a foregone conclusion.

- **Standing invitation:** the adversary is a lower bound. Reviewers running exact DP or POMDP solutions under sensing noise will widen these gaps; such results are welcome and strengthen the record.

6. Honest scope

All results are numerical simulations by the framework's author (numpy/scipy, seeded, deterministic given the noise schedule); code is public. This negative result is published unedited at its preregistered threshold, alongside the confirmations in Appendices A and D/E, per the program's standing rule that failures are reported as readily as successes.

"A false balance is an abomination to the LORD, but a just weight is His delight." — Proverbs 11:1